Z score for Outlier Detection - Python

Outlier detection is an important task in data as identifying outliers can help us to understand the data better and improve the accuracy of our models. One common technique for detecting outliers is **Z score**. It is a statistical measurement that describes how far a data point is from the mean, expressed in terms of **standard deviations**. It helps us to identify if a data point is relatively higher or lower than the mean and how far it deviates from the average value.

Z-Score Formula:

$$Z = \frac{x - \mu}{\sigma}$$

Z= X - μ / σ

Where:

- X = Value of the data point
- μ = Mean of the data
- σ = Standard deviation of the data

Interpretation of Z-Score

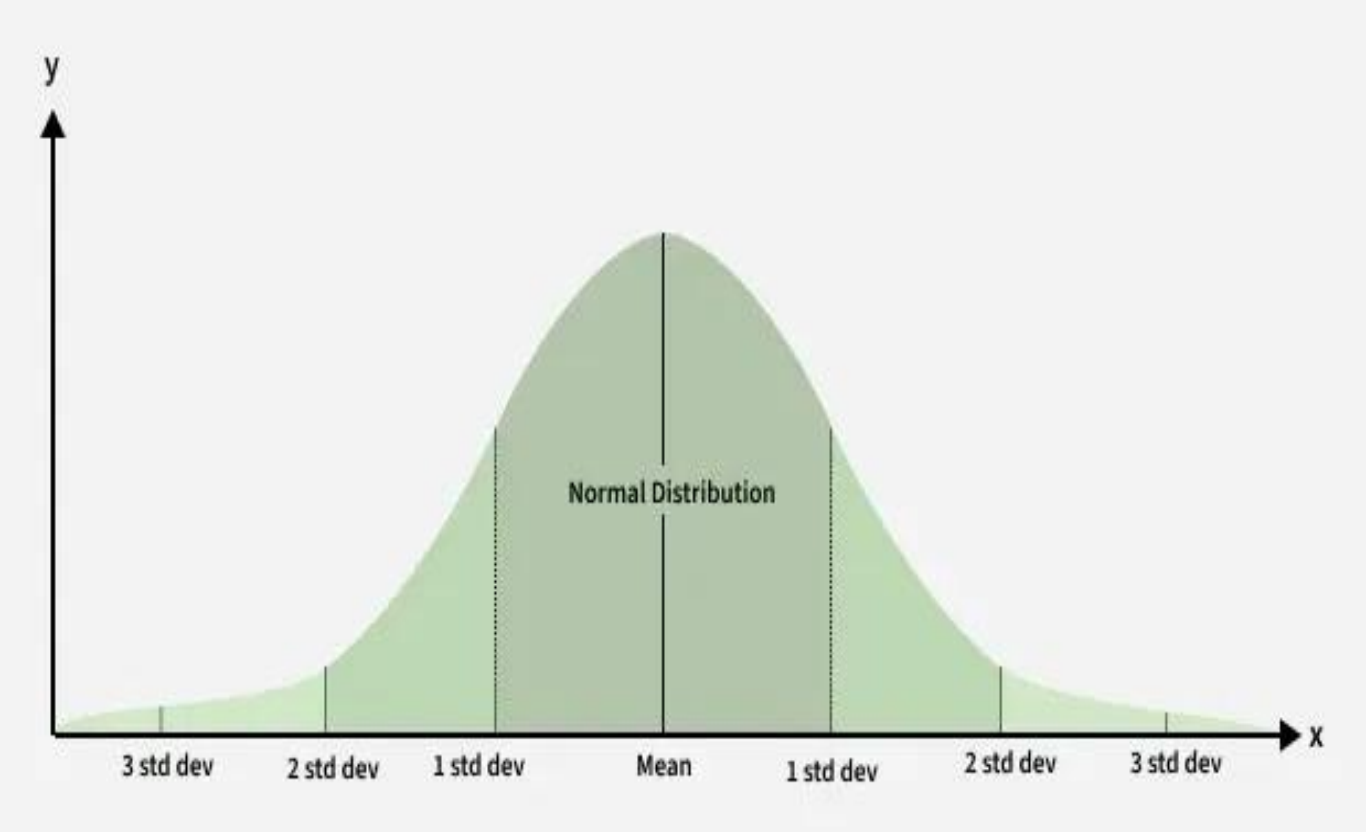
Outlier can be detected using Z-scores as follows:

- A **Z-score** of 0 shows that the data point is exactly at the mean.
- A **positive Z-score** shows that the data point is above the mean.
- A **negative Z-score** shows that the data point is below the mean.

Commonly, data points with a **Z-score greater than 3 or less than -3** are considered outliers as they lie more than 3 standard deviations away from the mean. This threshold can be adjusted based on the dataset and the specific needs of the analysis.

Why Use Z-Score for Outlier Detection?

Z-Score is useful for detecting outliers in **normal distribution**, as it highlights data points that lie far from the mean. A normal distributionis shown below and it is estimated that 68% of the data points lie between +/- 1 standard deviation. 95% of the data points lie between +/- 2 standard deviation 99.7% of the data points lie between +/- 3 standard deviation



For example, in a survey, it was asked how many children a person had. Suppose the data obtained from people is

1, 2, 2, 2, 3, 1, 1, 15, 2, 2, 2, 3, 1, 1, 2

Here, the value 15 is clearly an outlier as it deviates significantly from the other data points. The Z-Score for this data point will be much higher than the rest, showing it as an anomaly.

Lets see why it works so well:

- **Simple Calculation:** It's easy to calculate using just the mean and standard deviation.
- **Standardized Measure:** Z-Score standardizes the data which makes it easier to detect anomalies across different datasets with varying units or scales.
- **Identifying Extremes:** Find outliers which are the data points that lie far from the mean. Since Z-Score measures how far a data point is from the mean, it's perfect for identifying extremes in the data.

Steps for Z-Score Based Outlier Detection in Python

Let's see the steps of detecting outliers using the Z-Score method in Python.

Step 1: Importing Necessary Libraries

We will be importing numpy, pandas, scipy and matplotlib for calculating the Z-Score and visualizing the outliers.

```
import numpy as np

import pandas as pd

from scipy.stats import zscore

import matplotlib.pyplot as plt
```

Step 2: Creating the Dataset

For this example, we will use sample data and convert this into a pandas DataFrame.

```
data = [5, 2, 4.5, 4, 3, 2, 6, 20, 9, 2.5, 3.5, 4.75, 6.5, 2.5, 8, 1]
df = pd.DataFrame(data, columns=['Value'])
```

Step 3: Calculating the Z-Scores

Now, we calculate the Z-scores for this dataset using the z-score function from scipy.stats.

```
df['Z-score'] = zscore(df['Value'])
print(df)
```

Output:

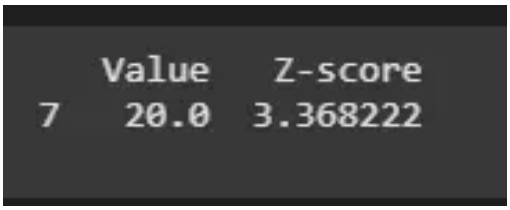
	Value	Z-score
0	5.00	-0.060721
1	2.00	-0.746510
2	4.50	-0.175019
3	4.00	-0.289317
4	3.00	-0.517913
5	2.00	-0.746510
6	6.00	0.167875
7	20.00	3.368222
8	9.00	0.853664
9	2.50	-0.632211
10	3.50	-0.403615
11	4.75	-0.117870
12	6.50	0.282173
13	2.50	-0.632211
14	8.00	0.625068
15	1.00	-0.975106

Step 4: Identifying Outliers

Next, we'll identify the data points that have a Z-score greater than 3 or less than -3 which are commonly considered outliers.

```
outliers = df[df['Z-score'].abs() > 3]
print(outliers)
```

Output:



Identifying Outliers

Step 5: Visualizing the Data

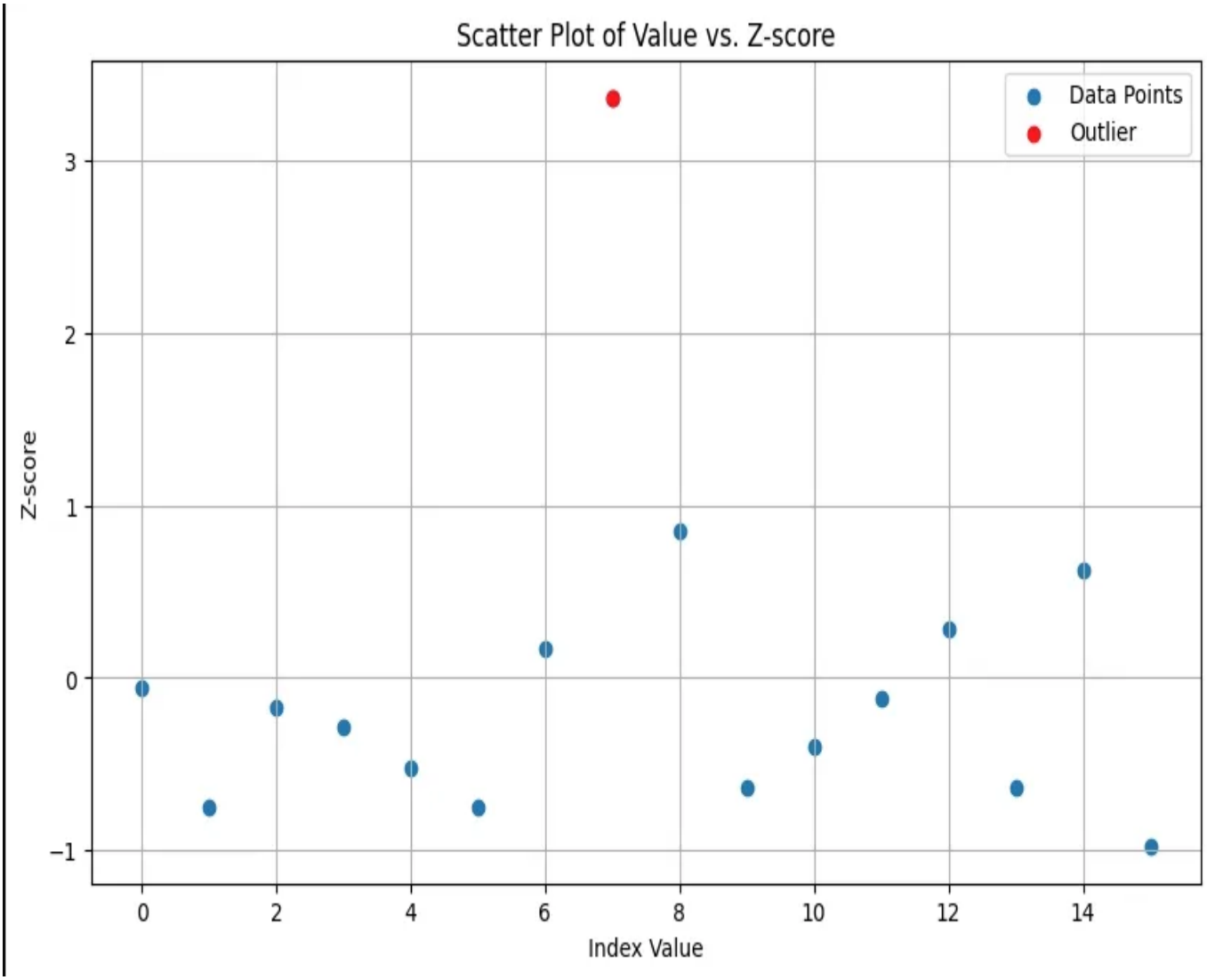
To better understand the outliers, let's create a scatter plot to visualize the dataset and highlight the outliers.

```
plt.figure(figsize=(10, 6))

plt.scatter(df['Value'].index, df['Z-score'], label='Data Points')
plt.scatter(outliers['Value'].index, outliers['Z-score'], color='red', label='Outlier')

plt.xlabel('Index Value')
plt.ylabel('Z-score')
plt.title('Scatter Plot of Value vs. Z-score')
plt.legend()
plt.grid(True)
plt.show()
```

Output:



Visualizing the Data

In this case, the value 20 is an outlier because its Z-score is significantly higher than the rest of the values in the dataset.

What is an outlier?

An outlier is a data point in a data set that is distant from all other observations. A data point that lies outside the overall distribution of the dataset.

What are the impacts of having outliers in a dataset?

- It causes various problems during our statistical analysis
- It may cause a significant impact on the mean and the standard deviation

What is the reason for an outlier to exists in a dataset?

- Variability in the data
- An experimental measurement error

What are the criteria to identify an outlier?

- Data point that falls outside of 1.5 times of an interquartile range above the 3rd quartile and below the 1st quartile
- Data point that falls outside of 3 standard deviations. we can use a z score and if the z score falls outside of 2 standard deviation

DIFFERENT OUTLIER DETECTION TECHNIQUE.

- Hypothesis Testing
- Z-score method
- Robust Z-score
- I.Q.R method
- Winsorization method(Percentile Capping)
- DBSCAN Clustering
- Isolation Forest
- Visualizing the data

In this Notebok, we will take a deep dive into our notebooks and learn how to detect outliers using *Z-Score*

Introduction To Z-Score¶

Z score is an important concept in statistics. Z score is also called standard score. This score helps to understand if a data value is greater or smaller than mean and how far away it is from the mean. More specifically, Z score tells how many standard deviations away a data point is from the mean.

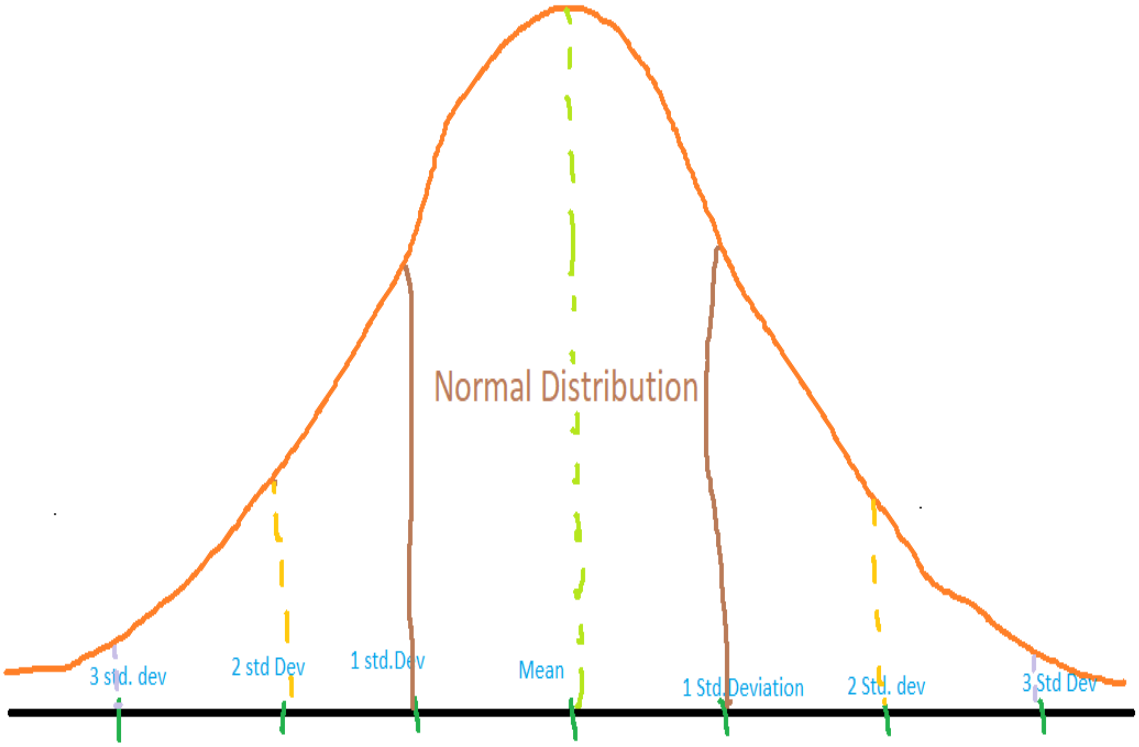
Formula For Z-Score

Formula for Z score = (Observation — Mean)/Standard Deviation

z = (X — μ) / σ

A normal distribution is shown below and it is estimated that¶

- ***68%* of the data points lie between +/- 1 standard deviation.**
- ***95%* of the data points lie between +/- 2 standard deviation**
- ***99.7%* of the data points lie between +/- 3 standard deviation**



Implementation of Z-Score in The Notebook

If the z score of a data point is more than 19, it indicates that the data point is quite different from the other data points. Such a data point can be an outlier.

dataset=
[11,10,12,14,12,15,14,13,15,102,12,14,17,13,15,10,15,12,10,14,13,15,10,19,107,10,13,12,14,12,108,12,11,14,]

Step 1: Import necessary libraries¶

```
import numpy as np
```

Step 2: Calculate mean, standard deviation

```
mean = np.mean(dataset)

std = np.std(dataset)

print('mean of the dataset is', mean)

print('std. deviation is', std)
```

mean of the dataset is 21.176470588235293

std. deviation is 26.372301186968762

Step 3: Calculate Z score¶

```
outliers=[]

def detect_outliers(data):


    threshold=3 # 3rd Standard Deviation

    mean = np.mean(data)

    std =np.std(data)



    for i in data:

        z_score= (i - mean)/std

        if np.abs(z_score) > threshold:

            outliers.append(i)

    return outliers

outlier_pt=detect_outliers(dataset)

outlier_pt
```

output : [102, 107, 108]

Conclusion

If you know the mean you know the standard deviation. Take your data point, subtract the mean from the data point, and then divide by your standard deviation. That gives you your Z-score.

You can use Z-Score to determine outliers. When you determine outliers it depends on you to delete them or use log, winsorize, and similar methods.